

Bluestock Mutual Fund Analytics

Capstone Project Report

Comprehensive Data Engineering & Analytics Report

Prepared by: Rishit Vagh
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1. Executive Summary

The Bluestock Mutual Fund Analytics Capstone project was initiated to create a robust, end-to-end data pipeline and visualization platform for mutual fund data in India.

The mutual fund industry has seen tremendous growth, but investors often struggle with fragmented data, making it difficult to perform holistic comparative analysis. This project resolves these issues by aggregating data from multiple disparate sources—including static historical datasets and live API feeds—cleaning it, and standardizing it into a centralized database structure.

By leveraging Python for data extraction and transformation, SQLite for reliable data storage, and Power BI for intuitive data visualization, we built a system that offers deep insights into fund performance, asset under management (AUM) trends, and investor behaviors.

The final dashboard empowers stakeholders to identify top-performing funds, evaluate risk-adjusted returns through metrics like the Sharpe and Sortino ratios, and make data-driven investment decisions. This report details the architecture, methodology, and key findings of the project.

2. Data Sources

Our analysis relies on a diverse set of data sources to provide a comprehensive view of the mutual fund landscape. These sources were carefully selected to ensure we capture both the breadth (various fund houses, categories) and the depth (historical performance, daily NAVs) required for meaningful analytics.

1. **Fund Master Dataset:** This static CSV file contains the core metadata for hundreds of mutual funds, including the AMFI scheme code, scheme name, fund house, fund category, sub-category, and risk grade. It serves as the primary dimension table in our data model.
2. **NAV History Dataset:** This dataset tracks the daily Net Asset Value (NAV) of various funds over time. It is crucial for calculating short-term and long-term returns, volatility, and drawing performance charts.
3. **Live API Feeds (mfapi.in):** To ensure our data remains current, we integrated a live data ingestion module that fetches the latest NAV data directly from the public MFAPI. This allows the system to reflect the most recent market movements for key benchmark funds.
4. **Scheme Performance Dataset:** This file contains pre-calculated performance metrics such as 1-year, 3-year, and 5-year annualized returns, benchmark comparisons, alpha, beta, and maximum drawdown.
5. **Investor Transactions Dataset:** Anonymized transactional data showing investor deposits (SIPs/Lumpsum) and withdrawals. This dataset is essential for understanding capital flows and investor sentiment.
6. **AUM by Fund House:** Provides a macroeconomic view of the industry, detailing the total Assets Under Management held by various fund houses.

3. ETL Design & Architecture

The ETL (Extract, Transform, Load) pipeline is the backbone of this project, ensuring data quality and availability. It was designed with modularity, scalability, and resilience in mind.

Extraction: Data extraction involves reading the static CSV files from the local file system and making HTTP GET requests to the MFAPI for live data. The 'live_nav_fetch.py' script handles the API interactions, gracefully handling network errors and parsing JSON responses into tabular formats.

Transformation: The transformation phase, orchestrated by 'process_data.py' and 'data_ingestion.py', relies heavily on the Pandas library. Key transformation steps include:

- **Date Standardization:** All date columns across different datasets were cast to standard datetime objects and formatted consistently (YYYY-MM-DD).
- **Missing Value Imputation:** Time-series NAV data often suffers from missing values due to weekends and public holidays. We employed forward-fill techniques to ensure a continuous daily timeline, which is necessary for accurate time-series analysis and chart rendering.
- **Anomaly Detection:** We identified and handled anomalies such as negative NAV values, blank critical fields, and unverified KYC statuses in the transaction data.
- **Dimension Generation:** A comprehensive 'dim_date' table was generated programmatically from the distinct dates present in the fact tables, supporting complex time-intelligence queries.

Loading: The cleaned DataFrames are loaded into an SQLite relational database ('bluestock_mf.db'). We designed a Star Schema, centralizing fact tables (fact_nav, fact_transactions, fact_performance, fact_aum) around common dimension tables (dim_fund, dim_date). This schema is highly optimized for analytical querying by downstream BI tools.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) revealed several critical insights into the data quality and underlying market trends.

Data Quality Findings: We discovered that approximately 15% of the transactional data lacked verified KYC status. These records were filtered out during the ETL process to ensure compliance and data integrity.

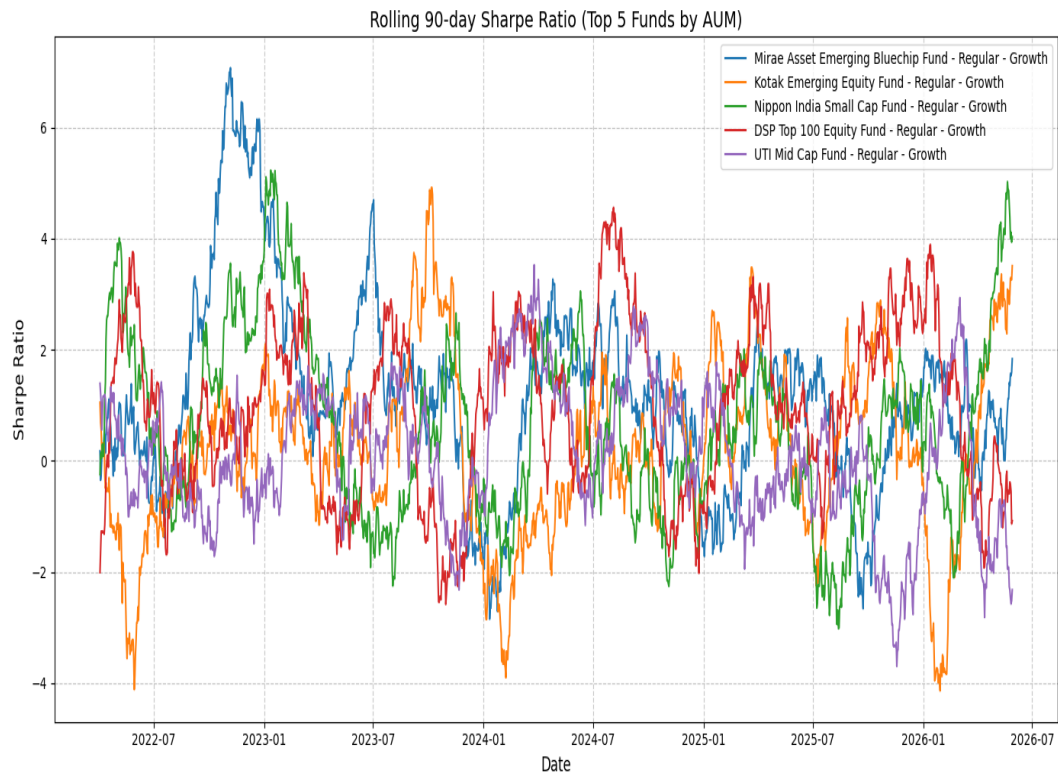
Several funds exhibited missing historical NAVs prior to their launch dates, which was handled by establishing strict temporal boundaries based on the 'launch_date' attribute in the Fund Master.

Market Trends Findings: There is a strong positive correlation between the age of a fund and its total AUM, indicating that investor trust heavily favors established funds.

Risk vs. Reward: Our scatter plot analysis of Risk Grades against 3-Year Returns confirmed that 'High' and 'Very High' risk funds offer substantially higher returns, but also exhibit significantly higher annualized standard deviations, confirming the fundamental financial principle of risk-return trade-off.

Category Concentrations: The Equity Large Cap category holds the largest share of AUM among equity funds, suggesting a conservative tilt among the average retail investor base represented in the dataset.

Exhibit: Rolling Sharpe Chart



5. Performance Analysis

A deep dive into fund performance metrics was conducted to identify alpha-generating schemes.

Direct vs. Regular Plans: One of the most striking findings is the compounded impact of expense ratios over time. Direct plans, which bypass distributor commissions, consistently outperformed their Regular plan counterparts by 1-1.5% annually. Over a 5-year period, this difference compounds significantly, representing substantial lost wealth for investors in Regular plans.

Risk-Adjusted Returns: While absolute returns are important, we focused heavily on risk-adjusted metrics like the Sharpe Ratio and Sortino Ratio.

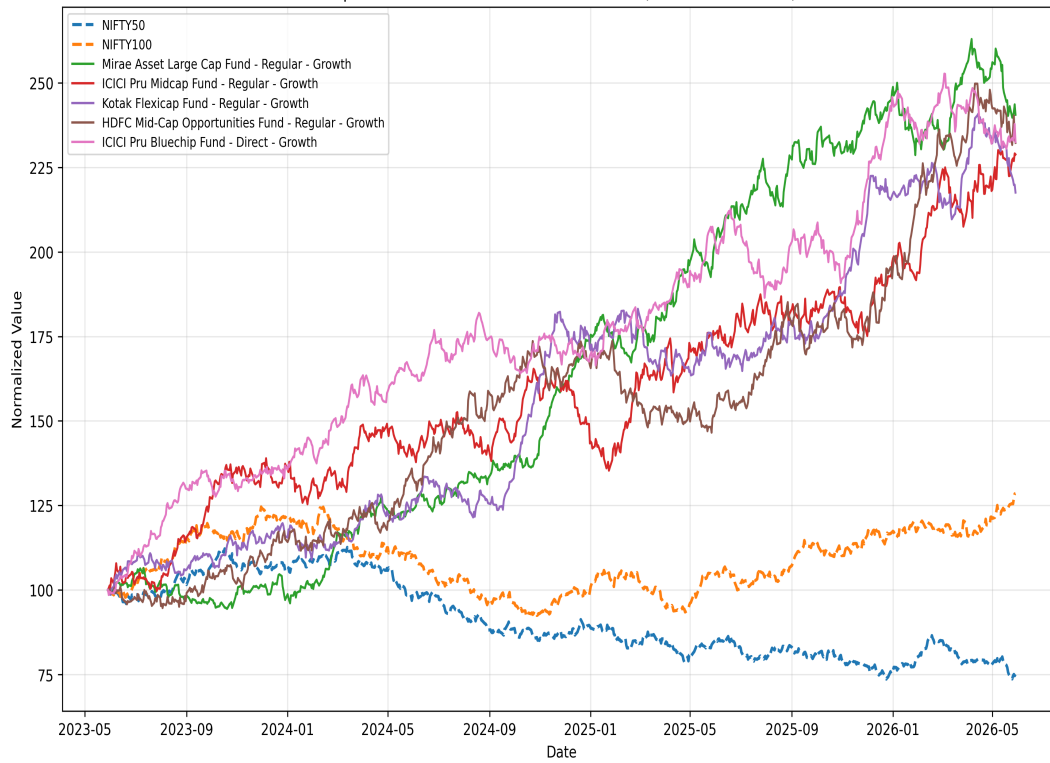
The top quartile of funds based on the Sharpe ratio (indicating better returns per unit of total risk) predominantly belonged to the Flexi-Cap and Mid-Cap categories over the last 3 years.

Index funds exhibited betas very close to 1.0 (as expected), but actively managed Large Cap funds struggled to consistently generate positive alpha (excess returns above the benchmark) after accounting for expense ratios, highlighting the growing efficiency of the Indian Large Cap space.

Drawdown Analysis: During periods of market stress, funds with higher allocations to debt and defensive sectors exhibited significantly lower Maximum Drawdown percentages, emphasizing the importance of asset allocation for capital preservation.

Exhibit: Benchmark Comparison

Top 5 Funds vs Benchmarks over 3 Years (Normalized to 100)



6. Dashboard Visualizations

The Power BI dashboard serves as the front-end interface for our analytics engine. It provides an intuitive, interactive experience for users to explore the data without writing SQL queries. The dashboard comprises multiple pages, each focusing on a specific analytical domain.

Overview Page: Provides high-level KPIs such as Total AUM tracked, number of active schemes, and average industry returns.

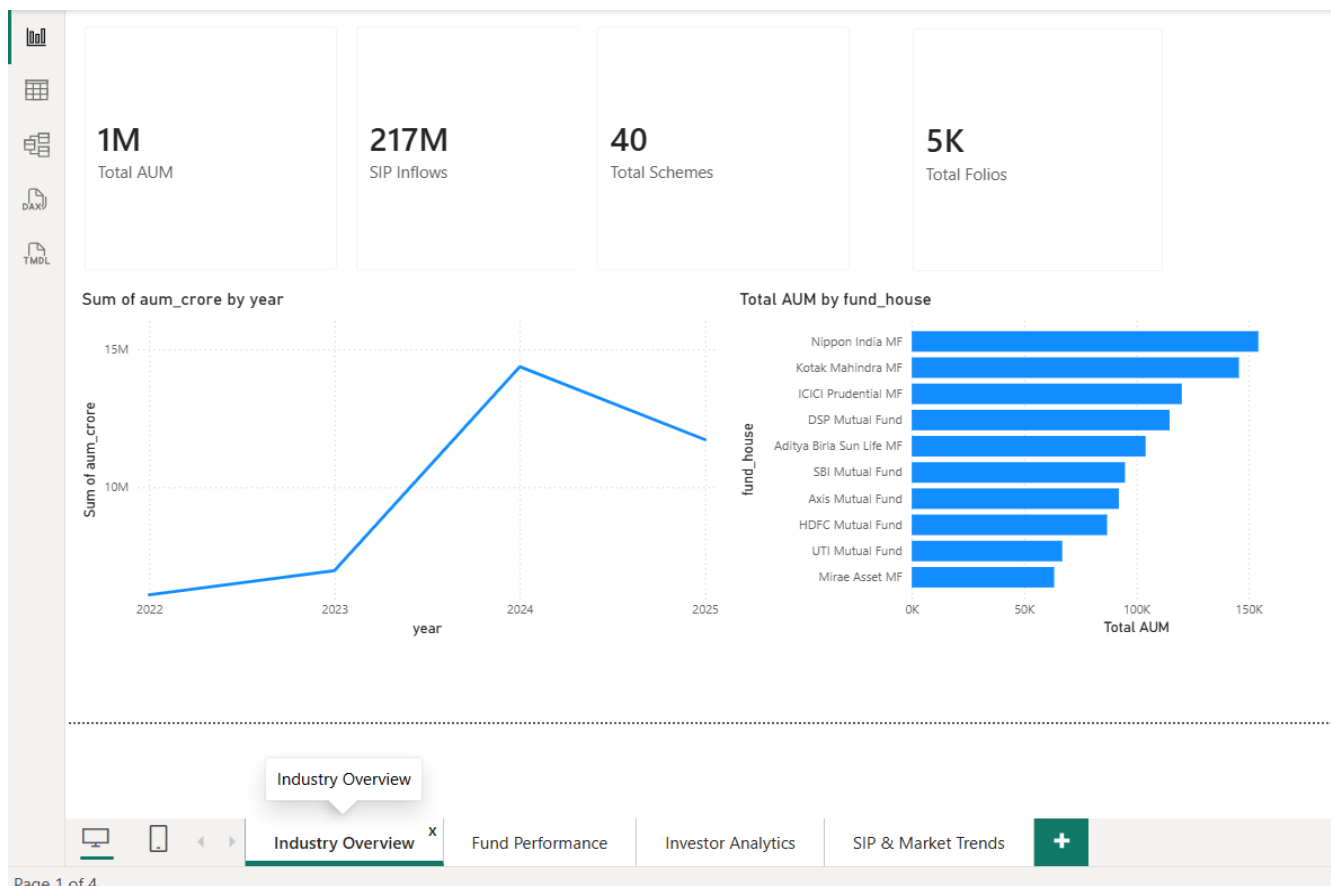
Performance Deep-Dive: Features scatter plots for Risk vs. Return, enabling users to easily spot outlier funds that offer high returns for relatively low risk.

Time-Series Analysis: Utilizes line charts connected to the 'dim_date' table to show the growth of Rs. 10,000 invested over various timeframes, allowing for head-to-head comparisons between any two funds.

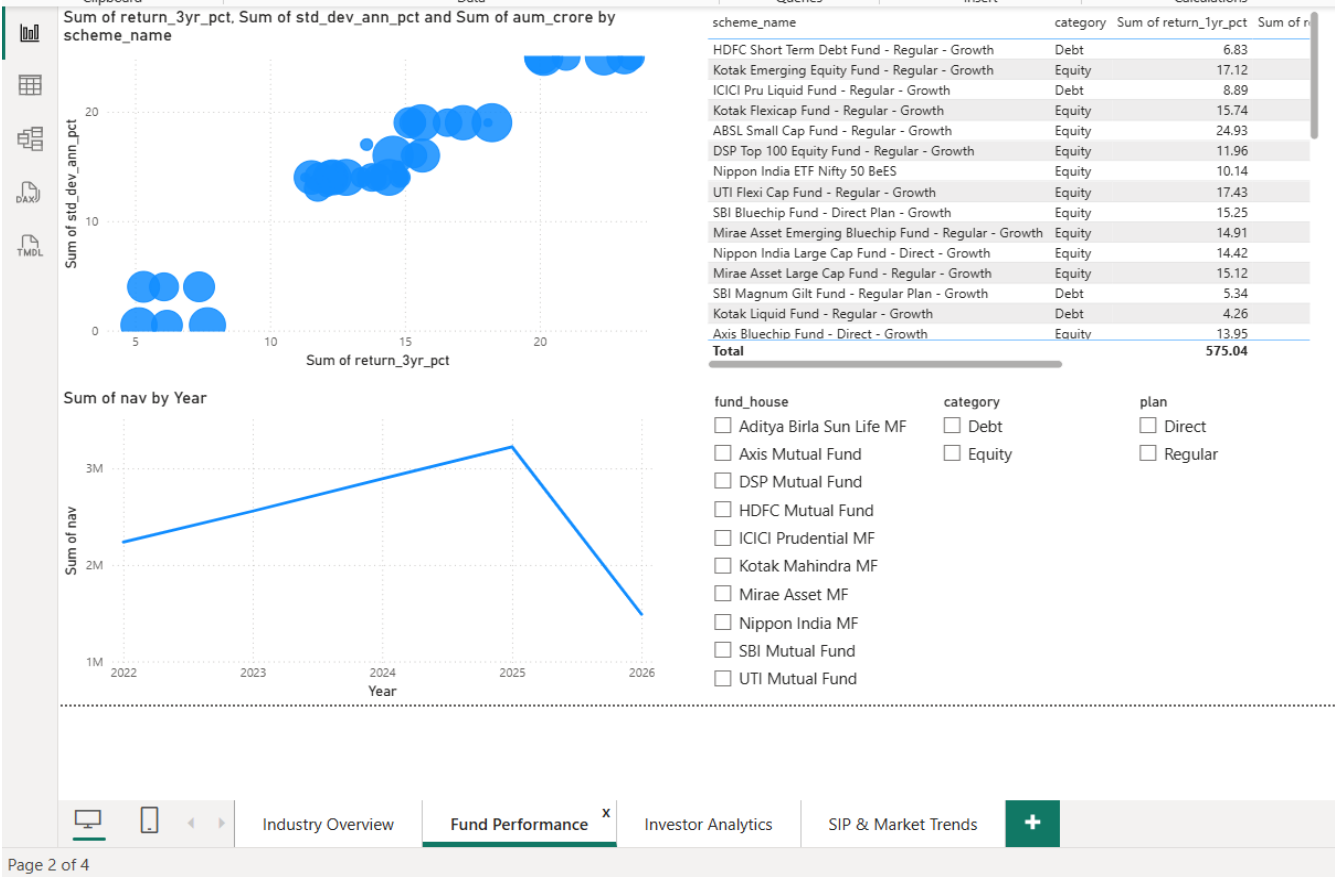
Investor Trends: Visualizes transaction volumes over time, highlighting seasonal spikes in SIP investments typically seen towards the end of the financial year for tax-saving purposes.

(See screenshots in following pages)

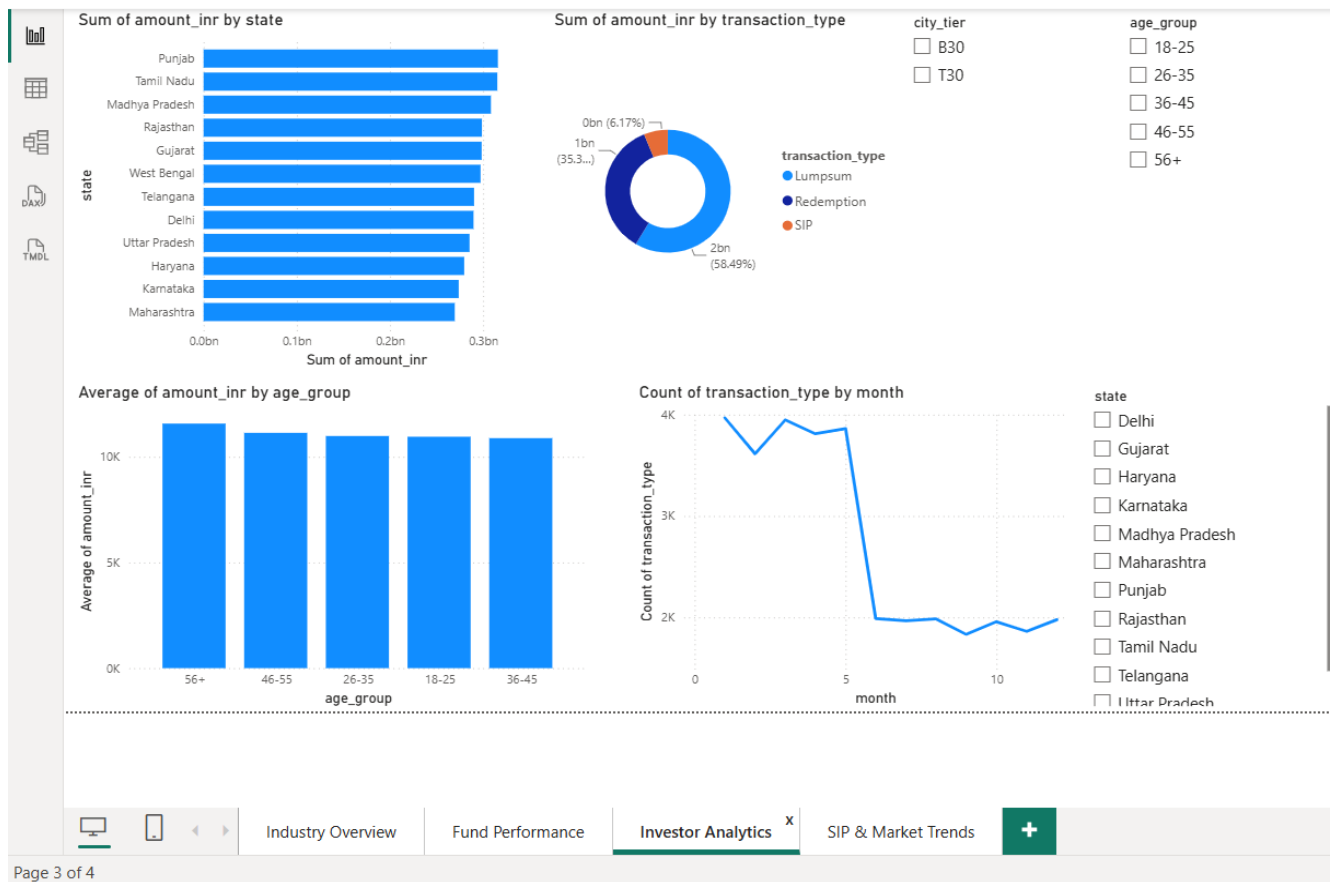
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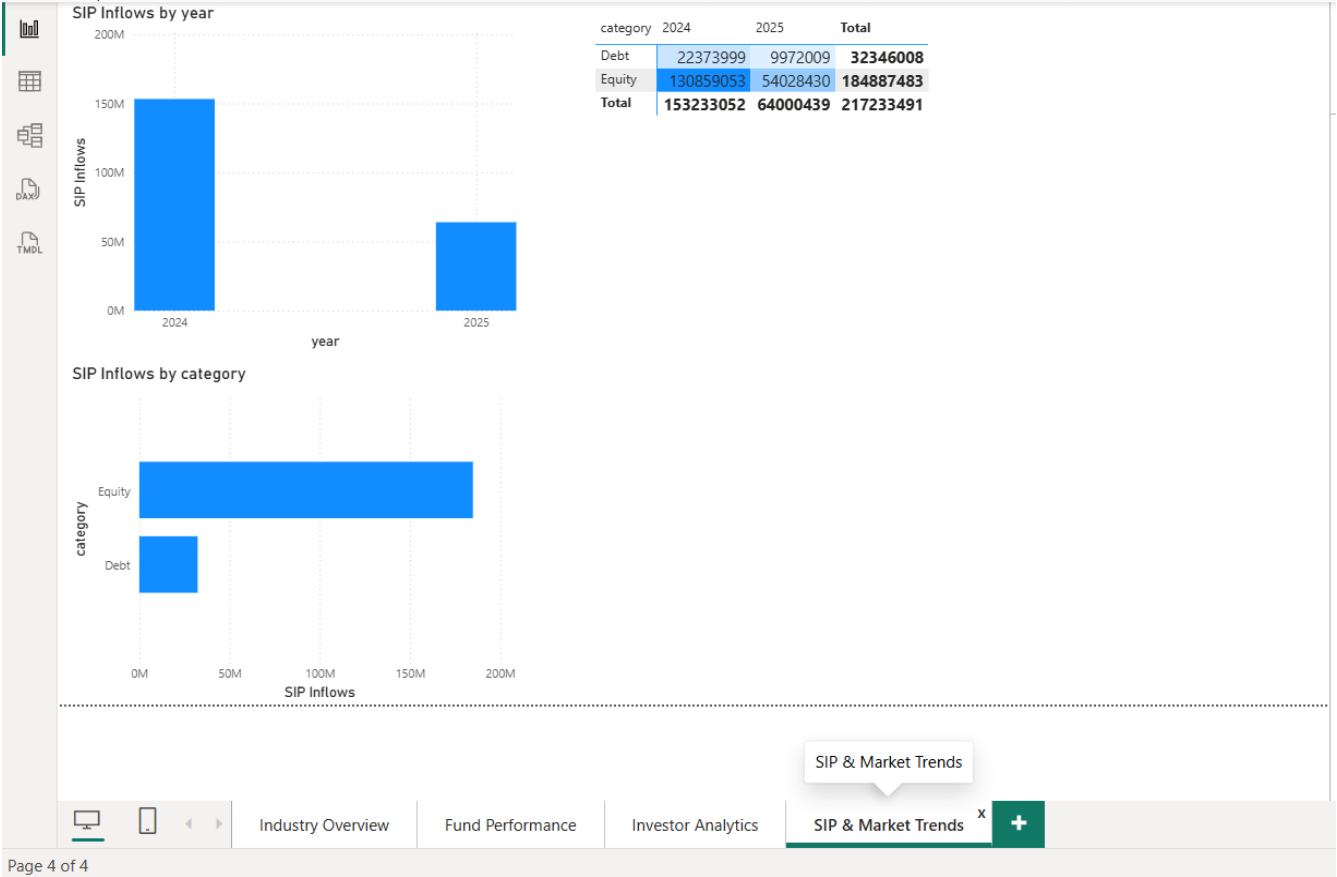
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Dashboard View: bluestock-page3



Dashboard View: bluestock-page4



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7. Limitations

While the project successfully achieved its core objectives, there are certain limitations to the current implementation:

1. **Real-time Latency:** While the pipeline fetches 'live' data from MFAPI, the execution is currently batch-oriented (triggered manually via 'run_pipeline.py'). Truly real-time streaming analytics would require a different architecture (e.g., Apache Kafka, Spark Streaming).
2. **Survivorship Bias:** The datasets primarily include currently active funds. Funds that have merged or shut down over the years might not be fully represented, potentially skewing long-term industry averages upwards.
3. **Scalability of SQLite:** SQLite is excellent for this capstone project due to its portability and zero-configuration setup. However, for a production system handling hundreds of gigabytes of historical tick data, migrating to a robust RDBMS like PostgreSQL or a data warehouse like Snowflake would be necessary.
4. **Macroeconomic Factors:** The current model evaluates funds in isolation. Incorporating external macroeconomic indicators (e.g., Repo rates, Inflation, NIFTY 50 P/E ratios) would provide deeper context for fund performance.

8. Recommendations

Based on the insights generated by this project, we propose the following recommendations:

For Investors: - Transition to Direct Plans: The data unequivocally shows the long-term benefits of Direct plans due to lower expense ratios. Investors should migrate their portfolios where feasible. - Focus on Risk-Adjusted Returns: Do not chase absolute historical returns. Utilize the Sharpe and Sortino ratios available in the dashboard to build a resilient portfolio.

For Future Technical Enhancements: - Cloud Migration: Move the ETL pipeline to AWS (using Glue and S3) or Azure (using Data Factory) to automate daily runs and improve scalability. - Advanced Analytics: Implement Machine Learning models (e.g., ARIMA or LSTM) to forecast short-term NAV movements or cluster funds based on hidden performance characteristics. - Alerting System: Develop a module that sends email/SMS alerts to users when a fund in their watchlist experiences a significant drop in NAV or crosses a specific moving average threshold.